HUSTmotor Multimodal Dataset

1. Dataset overview/数据集概览

This dataset comprises ***vibration*** signals and ***acoustic*** signals from motors in **6** different health states under **4** distinct operating conditions. These datasets are publicly available, and anyone can use them to validate diagnosis algorithms. **Publications making use of the *HUSTmotor multimodal dataset* is requested to cite the following paper.**

（Chao Zhao, Weiming Shen, Enrico Zio, Hui Ma, Multimodal unified generalization and translation network for intelligent fault diagnosis under dynamic environments. Engineering Applications of Artificial Intelligence, Volume 162, Part C, 2025, 112559.）

该数据集包含电机在 6 种不同健康状态和 4 种不同工况下的**振动信号**与**声学信号**。该数据集已公开，任何人都可以使用它来验证诊断算法。使用 *HUSTmotor multimodal dataset* 的出版物，请引用以下论文。

（Chao Zhao, Weiming Shen, Enrico Zio, Hui Ma, Multimodal unified generalization and translation network for intelligent fault diagnosis under dynamic environments. Engineering Applications of Artificial Intelligence, Volume 162, Part C, 2025, 112559.）

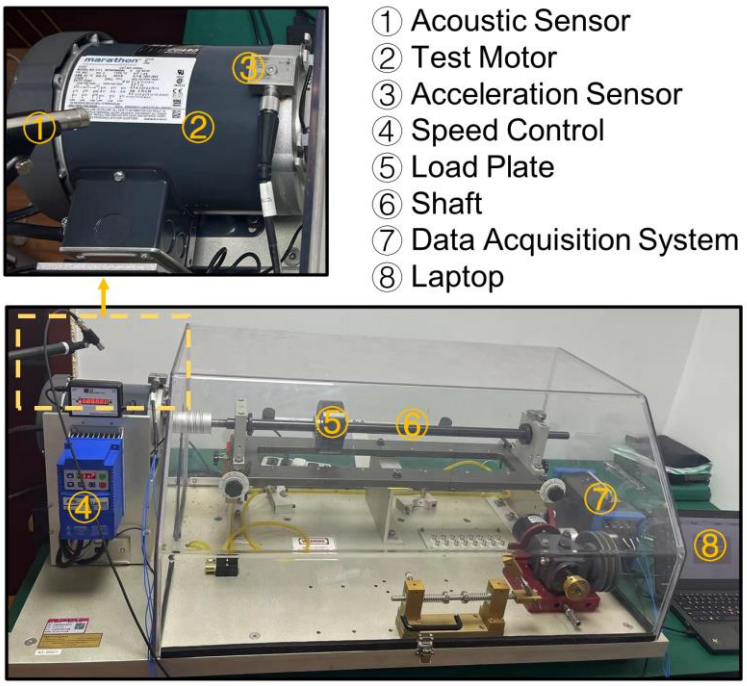


Fig. 1. (a) Test rig of HUSTmotor multimodal dataset.

2. Brief introduction to experiments/实验简介

2.1 Motor testbed/电机测试平台

The Motor fault tests were conducted using a Spectra-Quest Mechanical Fault Simulator, as depicted in **Fig. 1**. The model of the test motor is marathon, Cat. No. D393A.

电机故障测试使用 Spectra-Quest 机械故障模拟实验台进行，如图 1 所示。测试电机的型号是 marathon CAT. NO. D393A.

2.2 Motor health state/电机健康种类

The details of motors in six health states are presented in **Table 1**. All faults are artificially preset by the manufacturer. Therefore, schematic diagrams of the internal faulty components cannot be provided. Each faulty motor is independent, as shown in **Fig. 2**.

六种健康状态下电机的详细信息如表 1 所示。所有故障均由制造商人工预设，因此无法提供内部故障部件的示意图。每台故障电机都是独立的，如图 2 所示。

Table 1. Details of motor health states.

Index	Health state	Abbreviation
1	healthy	H
2	bearing fault	BF
3	bowed rotor	BOW
4	broken rotor bars	BROKEN
5	rotor misalignment	MISAL
6	voltage unbalance	UNBAL

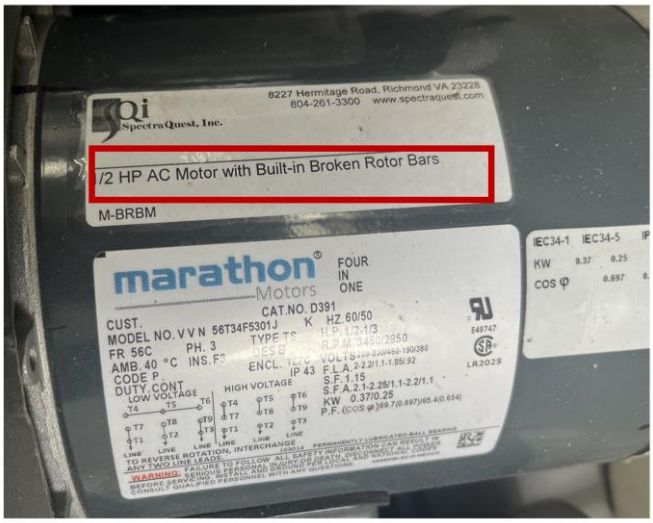


Fig. 2. Photograph of a motor with Bult-in Broken Rotor Bars.

2.3 Sensor information/传感器信息

The acceleration sensor used for data collection is shown in Fig. 3, with the model TREA331. The acoustic sensor is shown in Fig. 4, with the model AWA14004.

用于数据采集的加速度传感器如图 3 所示，型号为 TREA331。声学传感器如图 4 所示，型号为 AWA14004。

TREA331

LOW COST MINIATURE INDUSTRIAL TRIAXIAL ACCELEROMETER, SIDE EXIT 4 PIN MINI-MIL CONNECTOR, FOLLOWS CARTESIAN PHASE COORDINATE SYSTEM, 100 MV/G, ±15%

- 30-600,000 CPM Frequency Response (±3dB)
- -65 to 250°F (-54 to 121°C) Temperature Range
- Welded, Hermetic Sealing
- Collect 3 Channels of Data Simultaneously for Faster Data Collection
- Follows Cartesian Coordinate Phase Configuration (Right Hand Rule)

Fig. 3. Acceleration sensor type.

Fig. 4. Photographs of the acoustic sensor.

2.4 Working Condition/ 工况信息

A total of 4 different working conditions were set in experiments. The working conditions is shown in Tabel 2.

实验中共设置了 4 种不同的工况。工况如表 2 所示。

Tabel 2. Details of working conditions.

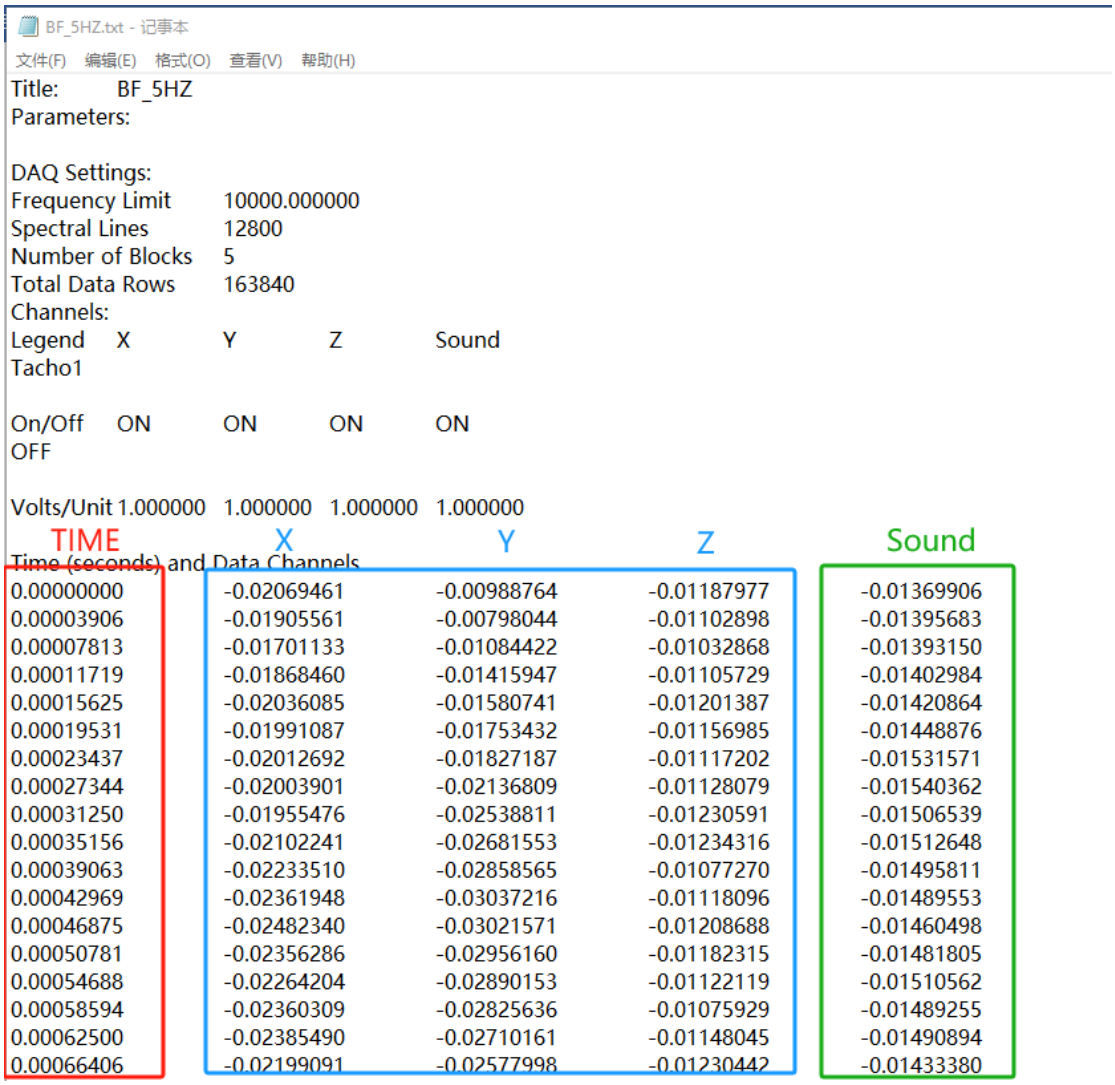
Index	Rotational speed (Hz)
Working condition 1	5
Working condition 2	10
Working condition 3	20
Working condition 4	30

2. 5 Data collection setting/ 数据采集

The sampling frequency is set to 25.6 kHz. As shown in Fig. 5, a total of 163,840 data points is recorded for each sampling (i.e., each sampling lasts 6.4 seconds). The dataset contains five columns: the first column represents time, columns 2, 3, and 4 correspond to the three-channel vibration signals, and the fifth column represents the acoustic signal.

采样频率设为 25.6 kHz。如图 5 所示，每次采样共记录 163840 个数据点（即每次采样时间持续 6.4 秒）。数据中共有 5 列，第一列为时间，第 2, 3, 4 列

分别三通道振动信号，第 5 列数据为声音信号。



BF_5HZ.txt - 记事本

文件(F) 编辑(E) 格式(O) 查看(V) 帮助(H)

Title: BF_5HZ
Parameters:

DAQ Settings:
Frequency Limit 10000.000000
Spectral Lines 12800
Number of Blocks 5
Total Data Rows 163840
Channels:

Legend	X	Y	Z	Sound
Tacho1				
On/Off	ON	ON	ON	ON
OFF				
Volts/Unit	1.000000	1.000000	1.000000	1.000000

TIME	X	Y	Z	Sound
Time (seconds) and Data Channels				
0.00000000	-0.02069461	-0.00988764	-0.01187977	-0.01369906
0.00003906	-0.01905561	-0.00798044	-0.01102898	-0.01395683
0.00007813	-0.01701133	-0.01084422	-0.01032868	-0.01393150
0.00011719	-0.01868460	-0.01415947	-0.01105729	-0.01402984
0.00015625	-0.02036085	-0.01580741	-0.01201387	-0.01420864
0.00019531	-0.01991087	-0.01753432	-0.01156985	-0.01448876
0.00023437	-0.02012692	-0.01827187	-0.01117202	-0.01531571
0.00027344	-0.02003901	-0.02136809	-0.01128079	-0.01540362
0.00031250	-0.01955476	-0.02538811	-0.01230591	-0.01506539
0.00035156	-0.02102241	-0.02681553	-0.01234316	-0.01512648
0.00039063	-0.02233510	-0.02858565	-0.01077270	-0.01495811
0.00042969	-0.02361948	-0.03037216	-0.01118096	-0.01489553
0.00046875	-0.02482340	-0.03021571	-0.01208688	-0.01460498
0.00050781	-0.02356286	-0.02956160	-0.01182315	-0.01481805
0.00054688	-0.02264204	-0.02890153	-0.01122119	-0.01510562
0.00058594	-0.02360309	-0.02825636	-0.01075929	-0.01489255
0.00062500	-0.02385490	-0.02710161	-0.01148045	-0.01490894
0.00066406	-0.02199091	-0.02577998	-0.01230442	-0.01433380

Fig. 5. Illustration of raw data file.

3. Dataset details/数据集细节

The raw data file comprises 24 files (6 health states multiplied by 4 working conditions), each in txt format. For instance, the filename "H_5HZ" indicates a healthy motor under the 5 Hz working condition. Abbreviation information is presented in **Table 2**.

原始数据文件共包含 24 个文件（6 种健康状态 × 4 种工况），每个文件为 txt 格式。例如，文件名“H_5HZ”表示在 5 Hz 工况下的健康（H）电机。简称请见**表 2**。

4. Data Visualization

Fig. 6 shows the visualization of signals for the healthy motor under the 30 Hz working condition.

图 6 是对健康状态电机在 30HZ 工况下的信号可视化图。

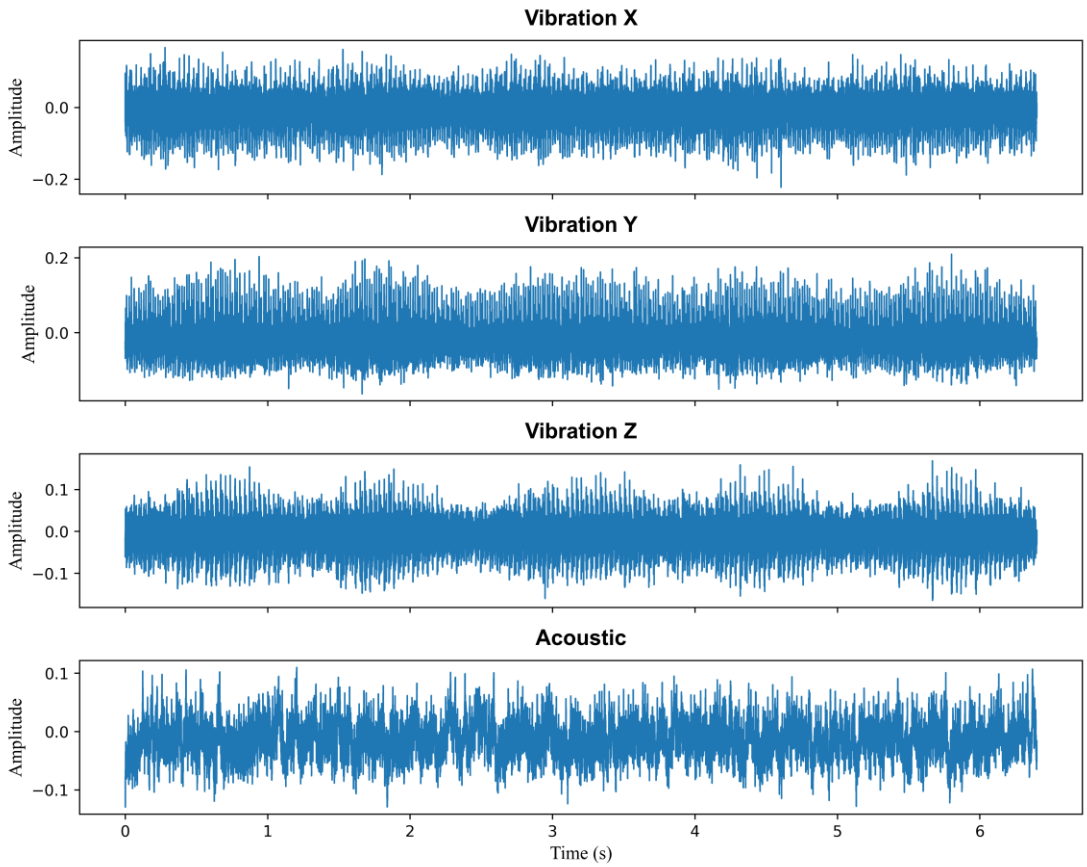


Fig. 6. Visualization of signals at 30 Hz in the healthy state

5. Contact/联系方式

✧ If you have any questions or suggestions, do not hesitate to contact:

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